

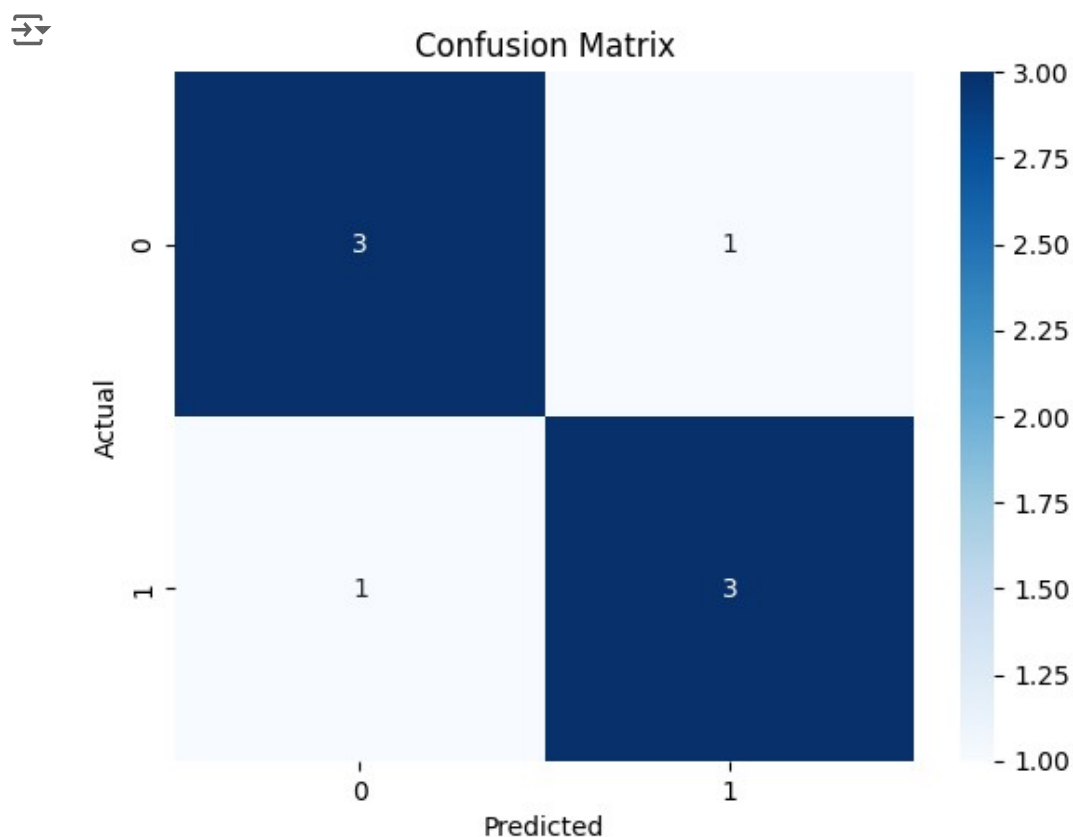
✓ Mastering TensorFlow: Day 12 - Model Evaluation and Metrics

```
import tensorflow as tf
import numpy as np
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Simulating predictions and true labels
y_true = [0, 1, 0, 1, 0, 1, 0, 1]
y_pred = [0, 1, 0, 0, 0, 1, 1, 1]

# Generating the confusion matrix
cm = confusion_matrix(y_true, y_pred)

# Visualizing the confusion matrix
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```



Explanation:

Confusion Matrix: A valuable tool for visualizing the performance of a classification model by comparing the predicted labels to the actual labels.

```
from sklearn.metrics import precision_score, recall_score, f1_score
```

```
# Calculating precision, recall, and F1 score
precision = precision_score(y_true, y_pred)
recall = recall_score(y_true, y_pred)
f1 = f1_score(y_true, y_pred)

print(f'Precision: {precision:.2f}')
print(f'Recall: {recall:.2f}')
print(f'F1 Score: {f1:.2f}')
```

```
➤ Precision: 0.75
  Recall: 0.75
  F1 Score: 0.75
```

Explanation:

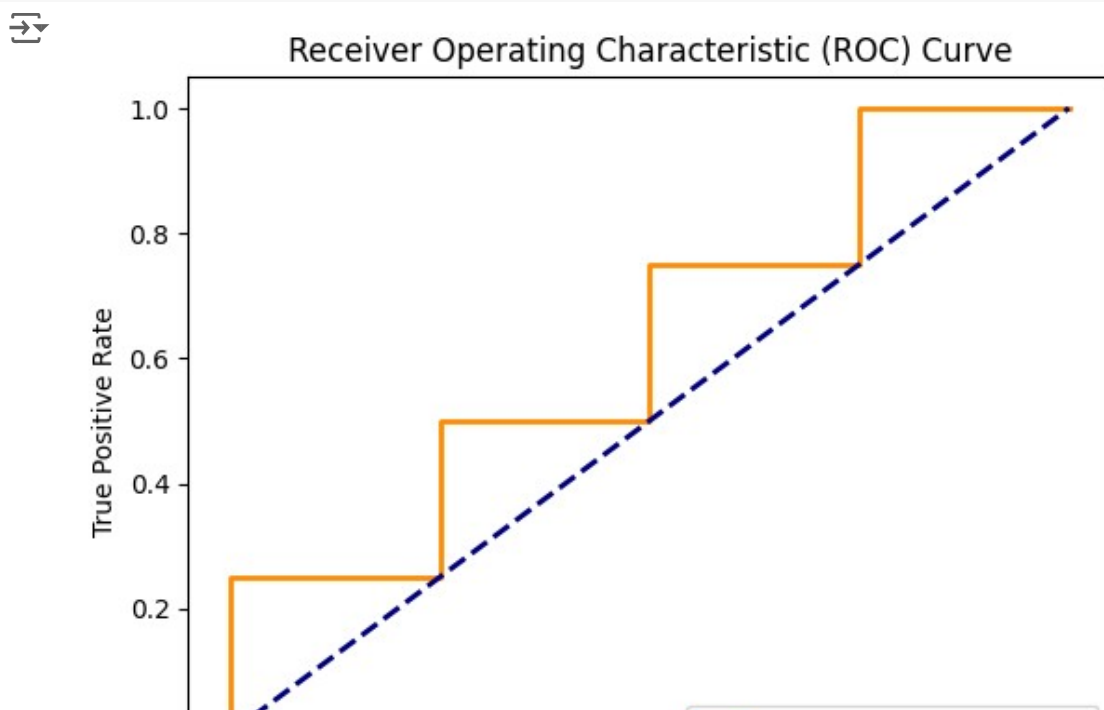
Precision, Recall, and F1-Score: These metrics provide insight into the trade-off between precision (how many selected items are relevant) and recall (how many relevant items are selected).

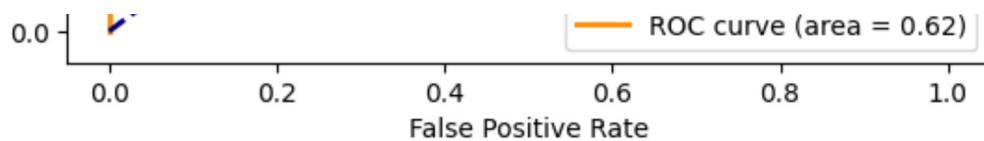
```
from sklearn.metrics import roc_curve, auc

# Simulating probability predictions
y_prob = [0.1, 0.4, 0.35, 0.8, 0.7, 0.2, 0.85, 0.9]

# Calculating ROC curve and AUC
fpr, tpr, _ = roc_curve(y_true, y_prob)
roc_auc = auc(fpr, tpr)

# Plotting the ROC curve
plt.figure()
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area = {roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc='lower right')
plt.show()
```





Explanation:

ROC Curve and AUC: These tools help assess the performance of binary classification models, particularly in scenarios where the data is imbalanced.

Practical Tips for Model Evaluation

1. **Confusion Matrix:** Always check the confusion matrix to identify which classes your model struggles with.
2. **Precision & Recall:** Balance these metrics based on the specific needs of your application.
3. **ROC & AUC:** Use the ROC curve and AUC to assess how well your model distinguishes between classes.

✓ Conclusion

Model evaluation is not just about accuracy. Using a combination of metrics like the confusion matrix, precision, recall, and ROC-AUC, you can get a much clearer picture of your model's performance. Today's session focused on these essential evaluation techniques, paving the way for better model tuning and optimization.

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